THEOREMS ON COMPLEXITY

PROF. ARTHUR M. HOBBS

Texas A & M University,
College Station, TX 77843

Section 1: Basic Definitions and Theorems

Because we want to allow both the domains of positive real numbers and positive integers, we will use D to stand for either Z^+ or $\mathcal{R}^+$. Further, we will use n for a variable which is to be restricted to Z^+, while x will be used interchangeably as a variable in $\mathcal{R}^+$ or a variable which may be in either domain. In a situation where we need to switch from a discrete variable to a continuous variable, as for example when we wish to take a derivative, as a shorthand for the process we will just switch from variable n to variable x. In the instance mentioned, this change of variable is to be understood as replacing the sentence, "Replace the function of a discrete variable by the function of a continuous variable with the same formula."

Definition 1. *Let $f(x)$ and $g(x)$ be functions from D to $\mathcal{R}$. We say "g* **dominates** *f," or "f* **is dominated by** *g," or "f* **is in big-oh of** *g" and write $f \in O(g)$ if there are constants k and m such that $|f(x)| \leq m|g(x)|$ for all $x \geq k$ when x is in the domain of f and g.*

Definition 2. *The set $O(g) = \{f | f \in O(g)\}$.*

Recall the triangle inequality for numbers a, b, and c: $|a+b| \leq |a| + |b|$. We use this in several proofs.

Example 1: Let $f(x) = x^2$ and $g(x) = x^2 + 2x - 5$. Then $|x^2| = x^2 \leq x^2 + 2x - 5 = |g(x)|$ if $x \geq 5/2$. Thus $f \in O(g)$ with $k = 5/2$ and $m = 1$. Further, $|g(x)| \leq x^2 + 2x + 5 \leq x^2 + 2x^2 + 5x^2 = 8x^2 = 8|x^2|$ if $x \geq 1$. Thus $g \in O(f)$ with $k = 1$ and $m = 8$.

Not all pairs of functions have a big-oh relationship. For example, for any k and m, there are values of $x \geq k$ for which $|x| > m|\tan x|$ and there are other values of $x \geq k$ for which $|\tan x| > m|x|$. So we restrict our attention to sets of functions within which all pairs f and g of functions do satisfy one of $f \in O(g)$ or $g \in O(f)$. We describe such a set of functions as a *set of comparable functions.*

One such set of comparable functions was introduced by G. H. Hardy[1]. This set is defined recursively as the smallest family $\mathcal{L}$ of functions satisfying all of

(1) The constant function $f(x) = \alpha$ is in $\mathcal{L}$ for every constant real number α.
(2) The identity function $f(x) = x$ for every x is in $\mathcal{L}$.
(3) If $f(x)$ and $g(x)$ are in $\mathcal{L}$, then $f(x) - g(x) \in \mathcal{L}$.
(4) If $f(x) \in \mathcal{L}$, then $e^{f(x)} \in \mathcal{L}$.

[1]G. H. Hardy, *Orders of Infinity: The 'Infinitärcalcül' of Paul du Boise-Reymond*, Cambridge University Press, 1910, second edition, 1924.

(5) If $f(x) \in \mathcal{L}$, and if there is an integer k such that $f(x) > 0$ for $x \geq k$, then $\log f(x) \in \mathcal{L}$.

It is possible to show that $\mathcal{L}$ includes all sums, products, quotients, and roots of functions in $\mathcal{L}$, and so it contains all polynomials, as well as exponential and logarithmic functions. It also includes such functions as $\log \log x$, x^x, $(x^2+3x)/(x-1)$, $(x+e^x)^{-1}$, etc. This set is described more fully on pages 428 and 429 of R. L. Graham, D. E. Knuth, and O. Patashnik, *Concrete Mathematics*, Addison-Wesley Publ. Co., Reading, Mass., 1989.

Theorem 1. *If $r \geq 0$ and $s > r$ are real numbers, then $x^r \in O(x^s)$ and $x^s \notin O(x^r)$.*

Proof. We may suppose $x > 0$, so the absolute values are not needed. Then notice, if $x \geq 1$, that $x^s = x^{s-r}x^r > x^r$. Hence $x^r \in O(x^s)$ with $m = 1$ and $k = 1$. Further, let m be any positive real number, and let k be the positive real number such that $k^{s-r} = m$. If $x \geq k$, then $x^s = x^{s-r}x^r \geq mx^r$. Hence there are no real numbers k and m such that $x \geq k$ guarantees that $x^s \leq mx^r$. It follows that $x^s \notin O(x^r)$. □

Definition 3. *We say "f* **is in big-omega of** *g" and write $f \in \Omega(g)$ if $g \in O(f)$. Further, we say "f* **is in big-theta of** *g" and write $f \in \Theta(g)$ if $f \in O(g)$ and $f \in \Omega(g)$ simultaneously.*

As in the case of big-oh, we let $\Omega(g)$ be the set of functions f such that $f \in \Omega(g)$ and we let $\Theta(g)$ be the set of functions f such that $f \in \Theta(g)$. As consequences of the definitions, we have the following relationships:

$$f \in O(g) \text{ implies } g \in \Omega(f);$$

$$f \in O(g) \text{ and } g \notin O(f) \text{ imply } f \in O(g) - \Theta(g); \text{ and}$$

$$f \in O(g) \text{ and } g \in O(f) \text{ imply } f \in \Theta(g).$$

According to Theorem 1, then, if $r \geq 0$ and $s > r$ we have $x^r \in O(x^s) - \Theta(x^s)$ and $x^s \in \Omega(x^r) - \Theta(x^r)$. Further, by Example 1 and the observations in the previous paragraph, $\Theta(x^2) = \Theta(x^2 + 2x - 5)$.

Theorem 2. *If $f \in O(g)$ and $g \in O(h)$, then $f \in O(h)$.*

Proof. Since $f \in O(g)$ and $g \in O(h)$, there are constants k_1, m_1, k_2, and m_2 such that $|f(x)| \leq m_1|g(x)|$ if $x \geq k_1$ and $|g(x)| \leq m_2|h(x)|$ if $x \geq k_2$. Let $k = \max(k_1, k_2)$. Then if $x \geq k$, we have $|f(x)| \leq m_1|g(x)| \leq m_1m_2|h(x)|$. Hence $f \in O(h)$ with $k = \max(k_1, k_2)$ and $m = m_1m_2$. □

Theorem 3. *If $f \in \Theta(g)$, then $\Theta(f) = \Theta(g)$.*

Proof. Since $f \in \Theta(g)$, we have both $f \in O(g)$ and $f \in \Omega(g)$. Let $h \in \Theta(f)$. Then $h \in O(f)$, so $h \in O(g)$ by Theorem 2. But also, $h \in \Omega(f)$ and $f \in \Omega(g)$, so $f \in O(h)$ and $g \in O(f)$; hence $g \in O(h)$, or $h \in \Omega(g)$. Thus $\Theta(f) \subseteq \Theta(g)$ (1).

Since $f \in \Theta(g)$, we have that $f \in O(g)$ and $f \in \Omega(g)$. These reverse to the statements $g \in \Omega(f)$ and $g \in O(f)$, respectively, so we have that $g \in \Theta(f)$. But then $\Theta(g) \subseteq \Theta(f)$ (2) by the first part of this proof. Combining (1) and (2), $\Theta(f) = \Theta(g)$. □

Theorem 4. *If $\Theta(f) \cap \Theta(g) \neq \emptyset$, then $\Theta(f) = \Theta(g)$.*

Proof. Suppose $\Theta(f) \cap \Theta(g) \neq \emptyset$, and let $h \in \Theta(f) \cap \Theta(g)$. Then by Theorem 3,

$$\Theta(f) = \Theta(h) = \Theta(g).$$

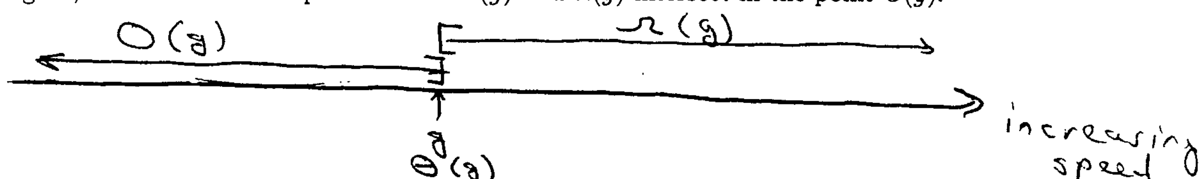
The theorem follows. □

According to this theorem, if we represent a class of comparable functions such as $\mathcal{L}$ by points on a straight line, the correct way to represent $\Theta(f)$ is by a point.

The idea of big-oh and its relatives is to give an approximate idea of how fast a function is growing as its variable tends to infinity. It is obvious that $x^2 + 2x - 5$ grows faster than x^2 does, but notice that

$$\lim_{x\to\infty} \frac{x^2}{x^2 + 2x - 5} = 1.$$

Thus in a very important sense, x^2 and $x^2 + 2x - 5$ grow toward infinity at the same rate. This is the sense we intend when we use the big-oh notation. With this in mind, let us imagine a line on which each point represents (in some precise sense) the rate of growth of a function as its variable tends to infinity. Then we can label to the point which represents the rate of growth of g with g and, according to Theorem 4, the set $\Theta(g)$ should be represented by the point g. Then $O(g)$ becomes the interval $(-\infty, g]$ and $\Omega(g)$ becomes the interval $[g, \infty)$. We obtain the following figure, in which the half-open intervals $O(g)$ and $\Omega(g)$ intersect in the point $\Theta(g)$.



Theorem 5. *Suppose f_1 and f_2 are both in $O(g)$. Then $f_1 + f_2 \in O(g)$.*

Proof. As in the proof of Theorem 2, since $f_1 \in O(g)$ and $f_2 \in O(g)$, there are constants k_1, m_1, k_2, and m_2 such that $|f_1(x)| \leq m_1|g(x)|$ if $x \geq k_1$ and $|f_2(x)| \leq m_2|g(x)|$ if $x \geq k_2$. Let $k = \max(k_1, k_2)$. Then $|f_1 + f_2| \leq |f_1| + |f_2| \leq m_1|g| + m_2|g| = (m_1 + m_2)|g|$. Thus $f_1 + f_2 \in O(g)$ with $k = \max(k_1, k_2)$ and $m = m_1 + m_2$. □

Theorem 6. *Let c be a non-zero constant. If $f(x) \in \Theta(cg(x))$, then $f(x) \in \Theta(g(x))$.*

Proof. As before, since $f(x) \in \Theta(cg(x))$, there are constants k' and m' such that $|f(x)| \leq m'|cg(x)|$ if $x \geq k'$. But then $|f(x)| \leq m'|cg(x)| = m'|c||g(x)|$. Thus $f(x) \in O(g(x))$ with $k = k'$ and $m = m'|c|$. But also, $cg(x) \in O(f(x))$, so there are constants $k"$ and $m"$ such that $|cg(x)| \leq m"|f(x)|$, whence $|g(x)| \leq (m"/|c|)|f(x)|$, so $g(x) \in O(f(x))$. Combining these, $f(x) \in \Theta(g(x))$. □

Theorem 7. *Let $f_1 \in O(g_1)$ and $f_2 \in O(g_2)$. Then $f_1 f_2 \in O(g_1 g_2)$.*

Proof. As in previous theorems, there are constants k_1, m_1, k_2, and m_2 such that $|f_1(x)| \leq m_1|g_1(x)|$ if $x \geq k_1$ and $|f_2(x)| \leq m_2|g_2(x)|$ if $x \geq k_2$. Let $k = \max(k_1, k_2)$. Then if $x \geq k$, we have $|f_1 f_2| = |f_1||f_2| \leq m_1|g_1|m_2|g_2| = (m_1 m_2)|g_1 g_2|$. The theorem follows. □

Corollary 7a. *Let $f_1 \in \Theta(g_1)$ and $f_2 \in \Theta(g_2)$. Then $f_1 f_2 \in \Theta(g_1 g_2)$.*

Proof. We apply the definition of Θ and Theorem 7 twice to get this result.

Intuitively, if $f \in O(g)$ and h is an increasing function, then we would expect that $h(f) \in O(h(g))$. This intuition is actually wrong, as shown by the case of $f(x) = 4x$, $g(x) = x$, and $h(x) = e^x$, since $e^{4x} \notin O(e^x)$ However, the next theorem provides that, under some rather technical conditions on the functions mentioned, this idea can work. We will see that many of the functions in $\mathcal{L}$ satisfy these conditions.

Theorem 8. *Suppose $f : D \to \mathcal{R}^+$, $g : D \to \mathcal{R}^+$ such that g is an increasing function, and suppose that $f \in O(g)$. Further, suppose b is a real number and $h : (b, \infty) \to \mathcal{R}^+$ such that h is increasing function and such that, for any positive constant c, there are constants d and N such that $h(cx) \le dh(x)$ whenever $x \ge N$. Suppose finally that, given N, there is a constant p such that $g(x) \ge N$ whenever $x \ge p$. Then $h(f(x)) \in O(h(g(x)))$.*

Proof. Since f and g are positive-valued functions, and since $f \in O(g)$, there is a number $c > 0$ and a number k_1 such that $f(x) \le cg(x)$ when $x \ge k_1$. But h is increasing, so there are positive constants d and N such that $h(f(x)) \le h(cg(x)) \le dh(g(x))$ when $g(x) \ge N$. But by hypothesis, there is a constant p such that $g(x) \ge N$ when $x \ge p$. Now we have $h(f(x)) \le dh(g(x))$ when $x \ge p$. Since h is positive-valued, it follows that $h(f(x)) \in O(h(g(x)))$ with $k = p$ and $m = d$. $\square$

We may notice that the last Theorem 8 condition on function g is satisfied if

$$\lim_{x \to \infty} g(x) = \infty.$$

As for h, we have the following:

Theorem 9. *The conditions of Theorem 8 on function h are satisfied for $h = \log x$ and for $h = x^r$ for any positive real number r.*

Proof. It is obvious that each of the named functions is positive-valued and increasing for $x \ge 1$. Thus we need only check that, for any positive constant c, there are constants d and N such that $h(cx) \le dh(x)$ whenever $x \ge N$. But $(cx)^r = c^r x^r$, so the condition in this case is satisfied for $d = c^r$ and $N = 1$. Similarly, $\log(cx) = \log c + \log x \le 2 \log x$ if $x \ge c$, and the condition is satisfied here with $d = 2$ and $N = c$. $\square$

For example, it is easy to show (we will do so in the next section) that $\log x \in O(x)$. It follows from Theorem 9 that $\log\log x \in O(\log x)$. Further, by Example 1 and Theorem 9, $(x^2 + 2x - 5)^{1/2} \in O(x)$.

Theorem 10. $n! \in O(n^n)$.

Proof. Since these functions are positive, we may omit the absolute values. Thus $n! = 1(2)(3)\ldots(n) \le n(n)(n)\ldots(n) = n^n$ for $n \ge 1$. Thus $n! \in O(n^n)$ with $k = m = 1$. $\square$

Corollary 10a. $\log(n!) \in O(n \log n)$, *or* $\sum_{i=1}^{n} \log i \in O(n \log n)$.

Section 2: Limit Theorems

We now connect the big-oh results to those of limits from earlier in your studies. These next few theorems will make finding the big-oh relation between many functions quite easy.

Theorem 11. *Suppose f and g are functions from Z^+ to $\mathcal{R}$, and suppose F and G are functions from $\mathcal{R}^+$ to $\mathcal{R}$ such that there is an integer N for which $F(n) = f(n)$ and $G(n) = g(n)$ for all integers $n \geq N$. Then $f(n) \in O(g(n))$ if $F(x) \in O(G(x))$.*

Proof. Since $F(x) \in O(G(x))$, there are constants m and k' such that $|F(x)| \leq m|G(x)|$ when $x \geq k'$. Let $k = \max(N, k')$. Then $|f(n)| = |F(n)| \leq m|G(n)| = m|g(n)|$ when $n \geq k$. Hence $f(n) \in O(g(n))$. □

Thus, for example, by Theorems 1 and 11, if $s > r \geq 0$, then $n^r \in O(n^s) - \Theta(n^s)$.

Theorem 12. *Suppose $f(x)$ and $g(x)$ are defined on $\mathcal{R}^+$ and suppose there is an N such that $g(x) \neq 0$ for all $x \geq N$. Then $f(x) \in O(g(x)) - \Theta(g(x))$ if*

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0.$$

Proof. Since

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0,$$

for any $m > 0$ there is a number $k \geq N$ such that $|(f(x)/g(x)) - 0| < m$ when $x > k$. But then $|f(x)| < m|g(x)|$ when $x > k$. Thus $f(x) \in O(g(x))$.

Now we wish to show that $g \notin O(f)$. Negating the definition of big-oh, we have that $g \notin O(f)$ if for all k' and for all $m' > 0$, there is a number x such that both $x > k'$ and $|g(x)| > m'|f(x)|$ hold. From the limit, since $|f(x)| < m|g(x)|$ when $x > k$ for any $m > 0$, we have $|g(x)| > (1/m)|f(x)|$ when $x > k$. But $m > 0$, so $1/m > 0$. Thus we have, for any $1/m > 0$, a number k such that both $x > k$ and $|g(x)| > (1/m)|f(x)|$. But given any real number k', we have both $x > \max(k, k')$ and $|g(x)| > (1/m)|f(x)|$. Thus $g \notin O(f)$. Thus $f \notin \Theta(g)$. This completes the proof. □

Inverting the fraction shows the following corollary:

Corollary 12a. *Suppose $f(x)$ and $g(x)$ are defined on $\mathcal{R}^+$ and suppose there is an N such that $f(x) \neq 0$ and $g(x) \neq 0$ for all $x \geq N$. Then $f(x) \in \Omega(g(x)) - \Theta(g(x))$ if*

$$\lim_{x\to\infty} \left|\frac{f(x)}{g(x)}\right| = \infty.$$

Theorem 13. *Suppose $f(x)$ and $g(x)$ are defined on $\mathcal{R}^+$ and suppose there is an N such that $g(x) \neq 0$ and $f(x) \neq 0$ for all $x \geq N$. Then $f(x) \in \Theta(g(x))$ if*

$$\lim_{x\to\infty} \left|\frac{f(x)}{g(x)}\right| = c,$$

for some constant $c > 0$.

Proof. Suppose

$$\lim_{x\to\infty} \left|\frac{f(x)}{g(x)}\right| = c$$

for constant $c > 0$. Then for every $\epsilon > 0$, there is a real number N such that $||f(x)/g(x)| - c| < \epsilon$ if $x > N$. Let $0 < \epsilon_0 < c$, so $c - \epsilon_0 > 0$. Then there is a number k such that $||f(x)/g(x)| - c| < \epsilon_0$ if $x > k$. Performing a little algebra, we have

$$-\epsilon_0 < \left|\frac{f(x)}{g(x)}\right| - c < \epsilon_0,$$

so

$$0 < c - \epsilon_0 < \left|\frac{f(x)}{g(x)}\right| < c + \epsilon_0.$$

Let $m = c + \epsilon_0$. Then

$$\left|\frac{f(x)}{g(x)}\right| < c + \epsilon_0 = m,$$

so $|f(x)| < m|g(x)|$ for $x > k$. Thus $f(x) \in O(g(x))$.

But

$$\lim_{x\to\infty}\left|\frac{f(x)}{g(x)}\right| = c$$

with $c > 0$ is equivalent to

$$\lim_{x\to\infty}\left|\frac{g(x)}{f(x)}\right| = 1/c.$$

Thus the argument before shows that $g(x) \in O(f(x))$. By the definition of Θ, it follows that $f(x) \in \Theta(g(x))$. $\square$

Summary Theorem. *Under reasonable conditions,*

$$\lim_{x\to\infty}\frac{f(x)}{g(x)} = 0 \textit{ implies } f(x) \in O(g(x)) - \Theta(g(x)),$$

$$\lim_{x\to\infty}\frac{f(x)}{g(x)} = c > 0 \textit{ implies } f(x) \in \Theta(g(x)),$$

and

$$\lim_{x\to\infty}\frac{f(x)}{g(x)} = \infty \textit{ implies } f(x) \in \Omega(g(x)) - \Theta(g(x)).$$

Let us use $f \prec g$ to mean that $f \in O(g) - \Theta(g)$. Then we have

Theorem 14. $1 \prec \log\log x \prec \log x \prec x^r \prec x^r \log x \prec a^x$ *for real numbers* $r > 0$ *and* $a > 1$. *Further, if* n *is an integer variable, then* $a^n \prec n!$. *Finally,*

$$\frac{1}{a^x} \prec \frac{1}{x^r\log x} \prec \frac{1}{x^r} \prec \frac{1}{\log x} \prec \frac{1}{\log\log x} \prec 1.$$

Proof. The last statement clearly follows from the first one. For examples of the first result, note that

$$\lim_{n\to\infty}\frac{a^n}{n!} = \lim_{n\to\infty}\prod_{i=1}^{n}\frac{a}{i}.$$

Let k be a fixed integer such that $a/k < 1$. Then

$$\prod_{i=1}^{k}\frac{a}{i} = c$$

is a constant, so

$$\lim_{n\to\infty}\frac{a^n}{n!} = c\lim_{n\to\infty}\prod_{i=k+1}^{n}\frac{a}{i} = 0.$$

Thus $a^n \prec n!$ by Theorem 12. Next, we prove $\log x \prec x^r$:

$$\lim_{x\to\infty}\frac{\log x}{x^r} = \lim_{x\to\infty}\frac{1/x}{rx^{r-1}} = \lim_{x\to\infty}\frac{1}{rx^r} = 0.$$

Thus this part of the theorem follows from Theorem 12. The other parts have similar proofs. □

In view of Theorems 12 and 13, it is very useful to have a collection of limits to infinity of quotients of functions which themselves tend to infinity. One particularly valuable limit of this type is the following:

Theorem 15. $\lim_{n\to\infty}\frac{(n!)^{1/n}}{n} = e^{-1}$.

Proof. The proof is very long and is not an appropriate part of these notes. □

Example 2: The following computations show how this theorem can be used to get interesting complexity relations between functions.

$$\lim_{n\to\infty}\frac{((2n)!)^{1/n}}{n^2} = 4\lim_{n\to\infty}\frac{((2n)!)^{1/n}}{4n^2} = 4\left(\left(\lim_{n\to\infty}\frac{((2n)!)^{1/n}}{4n^2}\right)^{1/2}\right)^2$$

$$= 4\left(\lim_{n\to\infty}\frac{((2n)!)^{1/2n}}{2n}\right)^2 = 4e^{-2}.$$

Since $4e^{-2}$ is a positive constant, $((2n)!)^{1/n} \in \Theta(n^2)$ by Theorem 13.

Example 3: Find a simplest function with the same complexity as the following function:

$$f(x) = (x^{\log\log x} + 1000e^{x\log x} + x^{1000})(x^e + 40x^2\log x).$$

Solution: By Theorems 5 and 7, it suffices to find the terms of each of the two factors of f that is simplest. For comparison purposes, recall that

$$x^{\log\log x} = e^{(\log\log x)\log x}.$$

Since $\log\log x \prec x$, there is a number k such that $\log\log x - x < 0$ for $x > k$. Hence

$$\lim_{x\to\infty}\frac{e^{(\log\log x)\log x}}{e^{x\log x}} = \lim_{x\to\infty} e^{((\log\log x)-x)\log x} = 0.$$

Thus

$$e^{(\log\log x)\log x} \prec e^{x\log x}$$

by Theorem 12. But

$$x^{1000} \prec e^{x\log x},$$

too, by Theorem 14. Further, by Theorem 6, we may ignore the constant multiplier in the second term of the first factor of f. Thus the first factor of f is dominated by

$$e^{x\log x}.$$

For the second factor, we may notice that $\log x \prec x^{e-2}$, and so $x^2\log x \prec x^e$. (The coefficient 40 may be ignored here as before, by Theorem 6.)

Combining the results for the two factors, we see that

$$f \in \Theta(e^{x\log x}x^e)$$

is the solution to the problem.

Section 3: Computing with Θ

In this section, we shall restrict ourselves to functions f having the property that there is a number k such that $f(x) > 0$ for all $x > k$. Let us call such functions **positive**. Notice that a positive function does not have to be positive for all values of x, but that it will be positive from some point on with increasing x. Throughout this section, the symbol k will be reserved for the smallest real number r such that all of the positive functions named in the definition or theorem are positive for all $x \geq r$.

Although the definitions and theorems in this section are stated and proved for Θ, with slight modifications they can be shown to be true for O and Ω.

The definition of big-O can be extended to big-Θ; since we are working with big-Θ in this section, this extension is useful. For the extension we need the notion of an unbounded set. A set A of real numbers or integers is *unbounded* if and only if for any finite interval $[a, b]$ of real numbers, there is an element of A not in $[a, b]$. Since a and b can be moved farther apart at will, this definition means that there are infinitely many elements of A outside of any finite interval. These elements may be mostly larger than b, mostly more negative than a, or both. Note that any positive function is non-zero on an unbounded set.

Theorem 16. *Let $g(x)$ be a function which is non-zero on an unbounded set. Then a function $f(x)$ is in $\Omega(g(x))$ if and only if there are numbers m and k' such that $|f(x)| \geq m|g(x)|$ for all $x \geq k'$.*

Proof. Since $f(x) \in \Omega(g(x))$, $g(x) \in O(f(x))$. Hence there are numbers k' and m' such that $|g(x)| \leq m'|f(x)|$ for all $x \geq k'$. But $g(x) \neq 0$ on an unbounded set, so $m' > 0$. Hence $|f(x)| \geq (1/m')|g(x)|$ for all $x \geq k'$. Letting $m = 1/m'$, the theorem follows. $\square$

Corollary 16a. *Let $f(x)$ and $g(x)$ be functions which are non-zero on an unbounded set. Then $f(x) \in \Theta(g(x))$ if and only if there are numbers m_1, m_2, and k' such that $m_1|g(x)| \leq |f(x)| \leq m_2|g(x)|$ for all $x \geq k'$.*

It often happens that we want to compute with complicated functions, but most of the complications are in terms dominated by other terms. It would be good to be able to simplify our computations by using big-Θ to "smooth out" the complications. For example, we would like to do the following computation without committing a mathematical *faux pas*.

$$
\begin{aligned}
(x^2&+2x + \Theta(\log x))(x^3 + x + \Theta(\log\log x)) \\
&= x^5 + 2x^4 + x^3 + 2x^2 + x^2\Theta(\log\log x) + 2x\Theta(\log\log x) + x^3\Theta(\log x) + x\Theta(\log x) \\
&= x^5 + 2x^4 + x^3 + 2x^2 + \Theta(x^2\log\log x) + \Theta(x\log\log x) + \Theta(x^3\log x) + \Theta(x\log x) \\
&= x^5 + 2x^4 + x^3 + 2x^2 + \Theta(x^3\log x) \\
&= x^5 + 2x^4 + \Theta(x^3\log x). \qquad \blacksquare
\end{aligned}
$$

Of course, much of this is clearly an abuse of notation, since we are adding functions to sets, but that is not the most serious problem. How can we be sure the computations involved are even *consistent*? The next definition and accompanying theorems and definitions explain the abuse of notation and assure us that the computations are correct.

Definition 4. *Given positive comparable functions $h(x)$ and $f(x)$ and a comparable function $g(x)$, we write $h(x) = f(x) + \Theta(g(x))$, or, when the domain is understood, $h = f + \Theta(g)$ if and only if there is a positive function $s(x) \in \Theta(g(x))$ such that $h(x) = f(x) + s(x)$.*

Definition 5. *Given positive comparable functions $h(x)$ and $f(x)$ and a comparable function $g(x)$, we write $h(x) = f(x)\Theta(g(x))$, or, when the domain is understood, $h = f\Theta(g)$ if and only if there is a positive function $s(x) \in \Theta(g(x))$ such that $h(x) = f(x)s(x)$.*

Thus, for example, $x^3 + 5x^2 + 7x + 10 = x^3 + \Theta(x^2) = x^3 + x^2\Theta(1)$.

Since in the following we will sometimes be using $\Theta(g)$ to represent a set and sometimes a function, to distinguish the two when it is not obvious, we will refer to the use of $\Theta(g)$ as a function as "being in the context of $t(x) + \Theta(u(x))$."

Theorem 17. *If $f(x)$ and $g(x)$ are positive comparable functions, then $f\Theta(g) \in \Theta(fg)$ in the context of $t(x) + \Theta(u(x))$.*

Proof. By definition, $f\Theta(g) = f(x)s(x)$ for some $s \in \Theta(g)$. Since $s \in \Theta(g)$, by Theorem 16 there are numbers m_1, m_2, and k' such that $m_1 g(x) \leq s(x) \leq m_2 g(x)$ for all $x \geq \max(k, k')$. Hence $m_1 f(x)g(x) \leq f(x)s(x) \leq m_2 f(x)g(x)$ for all $x \geq \max(k, k')$. Thus $fs \in \Theta(fg)$. □

Corollary 17a. *If $f(x)$ and $g(x)$ are positive comparable functions, then $\Theta(f\Theta(g)) = \Theta(fg)$ in the context of $t(x) + \Theta(u(x))$.*

Proof. This is immediate from Theorems 17 and 3. □

Theorem 17 and its corollary justify the following definition:

Definition 6. *If $f(x)$ and $g(x)$ are positive comparable functions, then $f\Theta(g) = \Theta(fg)$ in the context of $t(x) + \Theta(u(x))$.*

Theorem 18. *If $f(x)$ and $g(x)$ are positive comparable functions, then $\Theta(f) + \Theta(g) \in \Theta(f + g)$ in the context of $t(x) + \Theta(u(x))$.*

Proof. By definition, in the context of $t(x) + \Theta(u(x))$, $\Theta(f) + \Theta(g) = s(x) + t(x)$ for some positive functions $s \in \Theta(f)$ and $t \in \Theta(g)$. Since $s \in \Theta(f)$, by Theorem 16 there are numbers ℓ_1, ℓ_2, and k' such that $\ell_1 f(x) \leq s(x) \leq \ell_2 f(x)$ for all $x \geq \max(k, k')$. Similarly, since $t \in \Theta(g)$, by Theorem 16 there are numbers m_1, m_2, and k'' such that $m_1 g(x) \leq t(x) \leq m_2 g(x)$ for all $x \geq \max(k, k'')$. Thus, if $x \geq \max(k, k', k'')$, then

$$s(x) + t(x) \leq \ell_2 f(x) + m_2 g(x) \leq (\max(\ell_2, m_2))(f(x) + g(x)).$$

Moreover,

$$s(x) + t(x) \geq \ell_1 f(x) + m_1 g(x) \geq (\min(\ell_1, m_1))(f(x) + g(x)).$$

Hence $s + t \in \Theta(f + g)$ and the theorem follows. □

Corollary 18a. *If $f(x)$ and $g(x)$ are positive comparable functions, then $\Theta(\Theta(f) + \Theta(g)) = \Theta(fg)$ in the context of $t(x) + \Theta(u(x))$.*

Proof. This is immediate from Theorems 18 and 3. □

Theorem 18 and its corollary justify the following definition:

Definition 7. *If $f(x)$ and $g(x)$ are positive comparable functions, then $\Theta(f) + \Theta(g) = \Theta(f+g)$ in the context of $t(x) + \Theta(u(x))$.*

Theorem 19. *If $f(x)$ and $g(x)$ are positive comparable functions, then $\Theta(f)\Theta(g) \in \Theta(fg)$ in the context of $t(x) + \Theta(u(x))$.*

Proof. By definition, $\Theta(f)\Theta(g) = s(x)t(x)$ for some functions $s \in \Theta(f)$ and $t \in \Theta(g)$. Since $s \in \Theta(f)$, by Theorem 16 there are there are numbers ℓ_1, ℓ_2, and k' such that $\ell_1 f(x) \le s(x) \le \ell_2 f(x)$ for all $x \ge \max(k, k')$. Similarly, since $t \in \Theta(g)$, by Theorem 16 there are numbers m_1, m_2, and k'' such that $m_1 g(x) \le t(x) \le m_2 g(x)$ for all $x \ge \max(k, k'')$. Thus, if $x \ge \max(k, k', k'')$, then

$$s(x)t(x) \le (\ell_2)(f(x))(m_2)g(x) = (\ell_2)(m_2)(f(x))(g(x)).$$

Moreover,

$$s(x)t(x) \ge (\ell_1)(f(x))(m_1)(g(x)) = (\ell_1)(m_1)(f(x))(g(x)).$$

Hence $st \in \Theta(fg)$ and the theorem follows. $\square$

Corollary 19a. *If $f(x)$ and $g(x)$ are positive comparable functions, then $\Theta(\Theta(f)\Theta(g)) = \Theta(fg)$ in the context of $t(x) + \Theta(u(x))$.*

Proof. This is immediate from Theorems 19 and 3. $\square$

Theorem 19 and its corollary justify the following definition:

Definition 8. *If $f(x)$ and $g(x)$ are positive comparable functions, then $\Theta(f)\Theta(g) = \Theta(fg)$ in the context of $t(x) + \Theta(u(x))$.*

Theorem 20. *If $f(x)$ and $g(x)$ are positive comparable functions, and if $f \in O(g)$, then $f + \Theta(g) \in \Theta(g)$ in the context of $t(x) + \Theta(u(x))$.*

Proof. By definition, $f + \Theta(g) = f(x) + s(x)$ for some $s \in \Theta(g)$. Since $f \in O(g)$, there are numbers ℓ and k' such that $f(x) \le \ell g(x)$ for all $x \ge \max(k, k')$. Moreover, because $s \in \Theta(g)$, by Theorem 16 there are numbers m_1, m_2, and k'' such that $m_1 g(x) \le s(x) \le m_2 g(x)$ for all $x \ge \max(k, k'')$. Thus, if $x \ge \max(k, k', k'')$, $0 + m_1 g(x) \le f(x) + s(x) \le \ell g(x) + m_2 g(x) \le \max(\ell, m_2) g(x)$. Thus $f + s \in \Theta(g)$. $\square$

Corollary 20a. *If $f(x)$ and $g(x)$ are positive comparable functions, and if $f \in O(g)$, then $\Theta(f + \Theta(g)) = \Theta(g)$ in the context of $t(x) + \Theta(u(x))$.*

Proof. This is immediate from Theorems 20 and 3. $\square$

Theorem 20 and its corollary justify the following definition:

Definition 9. *If $f(x)$ and $g(x)$ are positive comparable functions, if $f \in O(g)$, then $f + \Theta(g) = \Theta(g)$ in the context of $t(x) + \Theta(u(x))$.*

Example 3: Given the following functions f and g, find the two highest order terms of the product fg and a big-Θ estimate of the remainder of the product. For this,

$$f(x) = x^7 + 3x^3 + 19x + 21\sqrt{x}\log x + 8/x,$$

and

$$g(x) = x^6 + x + \sum_{k=1}^{5} \frac{k^2}{x^k}.$$

Solution: Letting $f = x^7 + 3x^3 + \Theta(x)$ and $g = x^6 + \Theta(x)$, we have

$$\begin{aligned} fg &= (x^7 + 3x^3 + \Theta(x))(x^6 + \Theta(x)) \\ &= x^{13} + 3x^9 + \Theta(x)(x^7 + 3x^3 + x^6 + \Theta(x)) \\ &= x^{13} + 3x^9 + \Theta(x^8) + \Theta(x^4) + \Theta(x^7) + \Theta(x)\Theta(x) \\ &= x^{13} + 3x^9 + \Theta(x^8) + \Theta(x^4) + \Theta(x^7) + \Theta(x^2) \\ &= x^{13} + 3x^9 + \Theta(x^8) \end{aligned}$$

Theorem 21. *For positive functions $f_i(n)$,*

$$\sum_{i=1}^{n} \Theta(f_i(n)) = \Theta\left(\sum_{i=1}^{n} f_i(n)\right).$$

Proof. This is a routine extension of Theorem 18 and its consequential definition. □

Example: We can use this theorem in conjunction with our earlier methods to get some very nice results. For example,

$$\sum_{k=1}^{n}(k^2 + \Theta(k)) = \frac{n(n+1)(2n+1)}{6} + \Theta\left(\sum_{k=1}^{n} k\right) = \frac{n^3}{3} + \Theta(n^2).$$

The latter simplification occurred because we know that $\Theta(n^2)$ includes all constant multiples of n^2 and all functions of the form n^2 plus polynomials of degree less than 2. Thus the additional information in the product $n(n+1)(2n+1)$ is of no value to us.

EXERCISES ON BIG-OH, BIG-Θ, AND BIG-Ω

ARTHUR M. HOBBS
TEXAS A & M UNIVERSITY, COLLEGE STATION, TX 77843

1. Find a simplest big-Θ form for the following functions for which n is in the positive integers and x is in the positive real numbers.
   (a) $f(n) = n^{3/2}(e^n \log n + n^{17} \log n + n)$.
   (b) $g(n) = n^2(\log(n^{73} \log n) + n \log n + n \log\log\log n)$.
   (c) $h(x) = (x^5 \log x + x^{5.1} + 10^{12}x^2)(e^x \log x + e^{x \log x} + e^3 x^{\log x})$.
   (d) $\ell(n) = (17n! + 2n^n)(e^{n!} + (2^n)! + \log n^{n!})$.
2. Prove that $1 \in O(\log\log x) - \Theta(\log\log x)$.
3. Prove that $x^r \in O(x^s) - \Theta(x^s)$ if $0 < r < s \in R$.
4. Prove that $a^x \in \Omega(x^r) - \Theta(x^r)$ if a and r are positive real numbers and $a > 1$.
5. Use limit theorems to prove that:
   (a) $x \log x \in \Theta(10^5 x \log(12x))$.
   (b) $(n!)^{1/n} \in \Theta(n)$.
   (c) $\log(x^4) \in O(x^{1/10}) - \Theta(x^{1/10})$.
6. Prove: $e^n \in O(n!) - \Theta(n!)$.
7. Let x be a continuous variable on the interval (k, ∞).

   (a) Prove that, if $f(x)$ and $g(x)$ are functions, if $f(x) \neq 0$ for large enough x, and if there is a positive constant c such that

   $$\lim_{x\to\infty} \frac{f(x)}{g(x)} = c,$$

   then there is a positive constant d such that

   $$\lim_{x\to\infty} \frac{g(x)}{f(x)} = d$$

   if $f \neq 0$ in the interval $[k, \infty)$.

   (b) Find a formula for d in part (a).
8. Simplify: $(x^3 + O(x^2))(x + O(1/x))$.
9. Simplify: $\sum_{k=1}^{n}(e^k + \Theta(k))$.
10. Simplify: $\sum_{k=1}^{n}(k^3 + \Theta(k))$.